FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.4: AREA OF POLYGONS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation task. You will write a structured response to the driving question using everything you have learned across all 8 lessons. Your goal is to show that you can connect the mathematical tools you developed — sine area rule, Heron's formula, parallelogram area, regular pentagon and hexagon area formulas — back to the real problem faced by the contractor in Kericho County. Read each section prompt carefully. Use the exemplar responses as a guide for the quality and depth expected. Write clearly, show your reasoning, and always connect your mathematics to the real-world context of the courtyard.

Section 1: Restate the Problem and Explain Why It Is Challenging	
In your own words, describe the situation at the Kericho community hall courtyard. Explain specifically WHY the contractor cannot simply use the familiar formula Area = ½ × base × height for the triangular sections, and why knowing only the side length is enough (or not enough) for the hexagonal border tiles. What mathematical challenge does this create?	The contractor in Kericho County needs to tile a courtyard floor that is divided into several different polygon shapes — triangular sections, a parallelogram stage area, and a border of regular hexagonal tiles. The challenge is that for the triangular sections, the contractor knows the lengths of two sides and the angle between them, but there is no right angle and no way to physically drop a perpendicular to measure the height. The usual formula, Area = ½ × base × height, requires a perpendicular height, so it cannot be applied directly. Without the height, you are stuck — unless you find another route. For the hexagonal border tiles, knowing only the side length is actually enough, because a regular hexagon can be divided into six equilateral triangles, and its area depends entirely on its side length. However, you still need the correct formula to use that side length correctly. The core mathematical challenge is: how do you calculate area accurately when a perpendicular height is hidden or inaccessible? This is exactly what the new formulas we developed in this unit solve.

Section 2: Explain the Mathematical Tools You Developed	
Describe at least FOUR area formulas you learned in this unit. For each formula, state: (a) what information you need to	During this unit, we developed four key formulas for calculating polygon areas without needing a measured perpendicular height: 1. Sine Area Rule for a Triangle: If you know two sides (a and b) and the included angle (C) between them, the area is given by Area = ½ab sin(C). You need: two side lengths and the angle between them. In the Kericho courtyard, each triangular chalk-line

use it, (b) the formula itself using correct mathematical notation, and (c) one sentence connecting it to a specific part of the Kericho courtyard. Show that you understand when and why each formula is used.	section has two measured sides and a known included angle, so this formula lets the contractor calculate every triangular area directly. 2. Heron's Formula: If you know all three sides of a triangle (a, b, c), first calculate the semi-perimeter $s = (a + b + c) \div 2$, then Area $= \sqrt{s(s-a)(s-b)(s-c)}$. You need: all three side lengths, but no angles and no height. This is useful as a cross-check for the triangular sections when all three side lengths can be taped but the angle is hard to measure precisely. 3. Area of a Parallelogram using the Sine Rule: If you know two adjacent sides (a and b) and the included angle (θ), then Area $= ab \sin(\theta)$. You need: two adjacent side lengths and the included angle. The parallelogram stage area of the courtyard has two measurable sides and a known corner angle, making this formula the right tool. 4. Area of a Regular Hexagon: If you know the side length (l) of each regular hexagonal border tile, then Area $= (3\sqrt{3} \div 2) \times l^2$. You need: only the side length of the hexagon. Since all the border tiles are identical regular hexagons and the contractor knows their side length, this formula lets you calculate the area of each tile and multiply by the number of tiles to find the total border area.

Section 3: Apply the Mathematics — Solve the Courtyard Problem	
Use the measurements below to calculate the areas of THREE different sections of the Kericho courtyard. Show ALL your working clearly, including the formula you chose, why you chose it, your substitution, and your final answer with correct units. Then add up the three areas to find the combined total. **Given measurements:** **– Triangle Section A: two sides of 6 m and 8 m, included angle = 55°** **– Parallelogram Stage: two adjacent sides of 10 m and 7 m, included angle = 65°** **– Hexagonal Border Tile: side length = 0.4 m, and there are 48 tiles in the border**	Triangle Section A — Sine Area Rule: I know two sides and the included angle, so I use Area $= \frac{1}{2}ab \sin(C)$. Area $= \frac{1}{2} \times 6 \times 8 \times \sin(55°)$ Area $= \frac{1}{2} \times 48 \times 0.8192$ Area $= 24 \times 0.8192$ Area ≈ 19.66 m² Parallelogram Stage — Parallelogram Sine Rule: I know two adjacent sides and the included angle, so I use Area $= ab \sin(\theta)$. Area $= 10 \times 7 \times \sin(65°)$ Area $= 70 \times 0.9063$ Area ≈ 63.44 m² Hexagonal Border Tiles — Regular Hexagon Formula: I know the side length of each tile, so I use Area of one hexagon $= (3\sqrt{3} \div 2) \times l^2$. Area of one tile $= (3 \times 1.7321 \div 2) \times (0.4)^2$ Area of one tile $= 2.5981 \times 0.16$ Area of one tile ≈ 0.4157 m² Total border tile area $= 48 \times 0.4157 \approx 19.95$ m² Combined Total Area: $19.66 + 63.44 + 19.95 = 103.05$ m²

	The contractor should order tiles to cover at least 103.05 m², and would be wise to add a small percentage extra for cuts and wastage.

Section 4: Connect Back to the Driving Question	
Now directly answer the driving question: How can we calculate the exact area of any polygon-shaped surface — even when we cannot measure its height — so that builders, farmers, and designers in our community never order too much or too little material? In your answer, explain the general principle that makes this possible, mention at least two different real-life contexts in Kenya (beyond the Kericho courtyard) where this matters, and explain the consequence of getting the area calculation wrong.	We can calculate the exact area of any polygon-shaped surface without measuring its height by using formulas that replace the hidden height with information we CAN measure — such as side lengths and angles. The general principle is that trigonometry (especially the sine function) allows us to express perpendicular height in terms of a side and an angle, which we can always measure with a tape and a protractor. Heron's formula takes this even further — it needs no angles at all, only the three side lengths. By decomposing any polygon into triangles and applying these formulas, we can find the exact area of any shape. This matters in many real Kenyan contexts. For example, a tea farmer in Kericho dividing a plot of land into triangular sections for different tea varieties needs the exact area of each section to know how many seedlings to buy — buying too few means gaps in the farm, buying too many wastes money. Similarly, a road construction team in the Rift Valley laying tarmac on a trapezoidal road section needs the precise area to order the correct volume of bitumen — overordering is expensive, underordering causes dangerous delays. Getting the calculation wrong has direct financial consequences: the Kericho contractor who overestimates the courtyard area will waste thousands of shillings on unused tiles, while underestimating means the project stalls and the community waits. Precision in area calculation is not just a mathematics exercise — it is a tool that protects people's money, time, and livelihoods.

Section 5: Reflect on Your Learning and Identify Limits	
Reflect honestly on what you now understand that you did not understand at the start of this unit. What was the most important idea that shifted your thinking? Also identify ONE limitation of the methods we used — is there any situation where even these formulas might not give a perfectly accurate answer in real life, and how might a contractor deal with that?	At the start of this unit, I believed that you could never calculate the area of a triangle without seeing or measuring its height. The most important idea that shifted my thinking was realising that the height is not truly 'missing' — it is hidden inside the relationship between the sides and the angle. When I saw that $h = b \sin(C)$ and that this could be substituted directly into the area formula, the whole unit clicked into place. Every formula we developed was a way of unlocking that hidden height using information that IS accessible in the real world. However, there is an important limitation. All of our formulas assume that the measurements of sides and angles are perfectly accurate. In real life, a tape measure pressed across uneven ground, or an angle measured with a basic protractor on a sloping surface, will carry small errors. These errors multiply through the calculation. For example, if the contractor's tape gives the side of a triangle as 8 m when it is actually 8.05 m, the final area will be slightly off. A professional contractor would manage this by using precise surveying instruments (such as a total station or a laser distance measurer), by taking multiple measurements and averaging them, and by always ordering a small buffer — typically 5–10% extra material — to account for measurement uncertainty, cutting waste, and breakage. Mathematics gives us the best possible answer from the data we have; good practice in the field means making sure the data is as accurate as possible.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of the Problem and the Mathematical Challenge	Clearly and accurately explains why perpendicular height is inaccessible in the Kericho courtyard scenario. Precisely identifies what information IS available (sides and angles) and articulates the mathematical gap this creates. Uses correct mathematical vocabulary (perpendicular height, included angle, polygon decomposition).	Explains the problem adequately and identifies the missing height as the core challenge. May not fully explain why the height is inaccessible or what specific measurements the contractor does have. Vocabulary is mostly correct.	Shows a basic awareness that the height is a problem but cannot explain why or connect it to the specific courtyard context. May confuse which measurements are available. Vocabulary is limited or imprecise.
Knowledge and Correct Application of Area Formulas	States all four required formulas correctly using proper mathematical notation. For each formula, correctly identifies the required inputs, demonstrates understanding of when to use it, and explicitly links it to the correct part of the courtyard. No errors in formula recall.	States at least three formulas correctly. Minor errors in notation or in identifying the required inputs for one formula. Links to the courtyard context are present but may be brief or partially incomplete.	States fewer than three formulas, or makes significant errors in one or more formulas (e.g., missing the ½ in the sine area rule, or using diameter instead of side length for the hexagon). Links to the courtyard are weak or absent.
Accuracy and Clarity of Mathematical Working	All three calculations are correct, fully shown step-by-step (formula stated → values substituted → arithmetic shown → answer with correct units). Final combined total is correct. Evidence of understanding WHY each formula was chosen for each section.	At least two of the three calculations are correct and reasonably well shown. Minor arithmetic errors may be present but the method is sound. Units are mostly correct. Formula choice is correct even if working is partially incomplete.	Only one calculation is fully correct, or significant errors in method (e.g., using wrong formula for a section, missing multiplication by number of tiles, omitting units). Working is difficult to follow or incomplete.
Connection to the Driving Question and Real-Life Contexts	Directly and fully answers the driving question. Articulates the general principle (trigonometry replaces hidden height) clearly. Provides two or more distinct, plausible Kenyan real-life contexts with specific details. Explains the real consequences of calculation errors in those contexts convincingly.	Answers the driving question adequately and mentions at least one Kenyan real-life context. The general principle is stated but may lack depth. Consequences of error are mentioned but not fully explored.	Partially answers the driving question or gives a very general answer not connected to Kenyan contexts. The general principle is not articulated. Real-life consequences of error are absent or very vague.
Quality of Reflection and Critical Thinking about Limitations	Identifies a specific, genuine conceptual shift in understanding (not just 'I learned new formulas'). Clearly articulates one real and relevant limitation of the methods (e.g., measurement error in the field). Proposes a practical, realistic strategy a contractor could use to manage this limitation. Reflection is honest and mathematically grounded.	Describes a meaningful learning shift and identifies a limitation, but the limitation may be somewhat generic (e.g., 'measurements might be wrong') without exploring why or how much this matters. The proposed strategy for managing the limitation is reasonable but brief.	Reflection is superficial (e.g., 'I found it hard but now I understand'). Limitation identified is trivial, incorrect, or not connected to the methods used. No strategy is proposed for managing the limitation in practice.